

How do CO₂ emissions compare between Europe and North America?

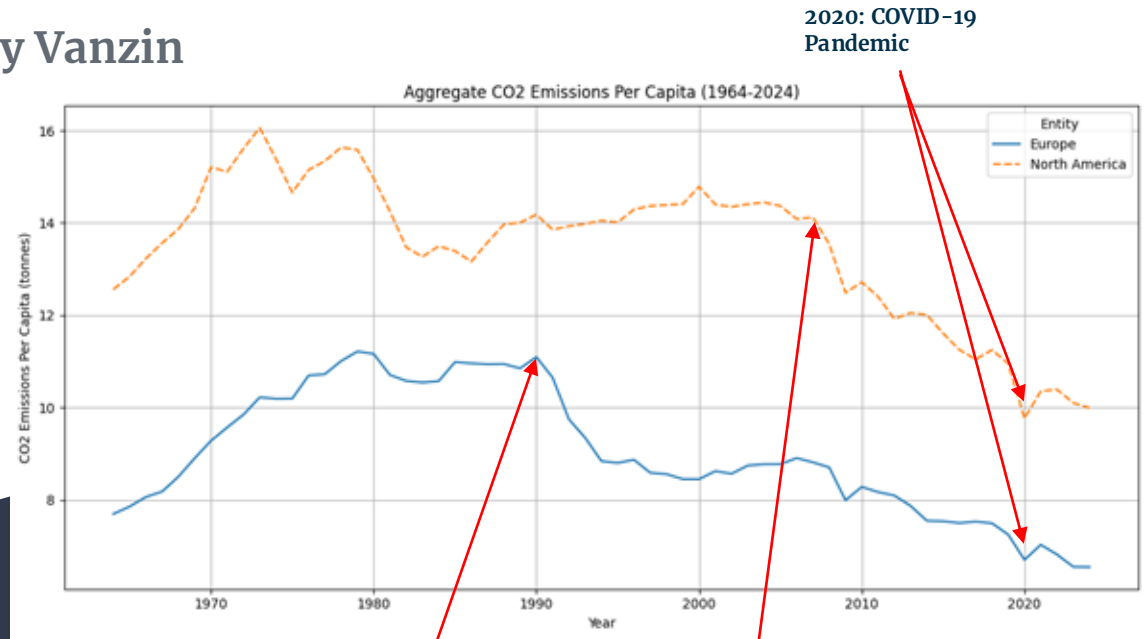
By Nolan Mikesell and Anthony Vanzin

H₀ (Null): $B_1 = 0$

- There is no statistically significant difference between the CO₂ emissions of Europe and North America over the last 60 years.

H_A (Alternative): $B_1 \neq 0$

- There is a statistically significant difference between the CO₂ emissions of Europe and North America over the last 60 years.

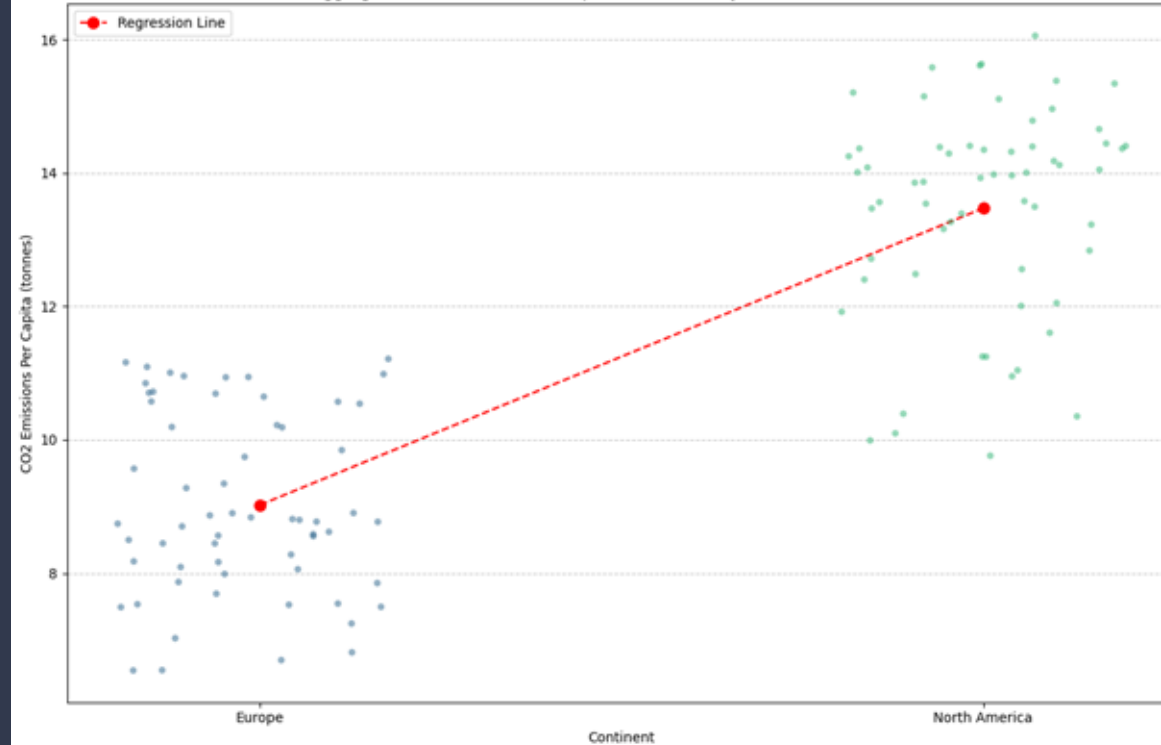


2020: COVID-19
Pandemic

1990: European Shift from Coal to
Natural Gas/Renewable Energy

2007: United States shift from Coal to
Natural Gas/Renewable Energy.

Aggregate CO2 Emissions Per Capita Distribution by Continent (1964-2024)



European Mean CO2 Emissions:
9.02 tonnes per capita.

North American Mean CO2
Emissions: **13.47 tonnes per capita.**

Regression Equation:

Emissions = 9.0232 + 4.449(Year)

- $B_0 = 9.0232$ (European μ)
- $B_1 = 4.449$ (Positive Slope)

P-value: **0.00001**

Conclusion: There is a statistically significant difference in mean CO2 emissions per capita between Europe and North America from the period 1964-2024.